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Hydrate mitigation in a pipeline with vortex tubes

Abstract

A system to prevent the formation of hydrates in a pipeline includes a heater assembly. The heater assembly has a vortex tube mounted on an outer surface of a first section of the pipeline and a compressed gas source. The vortex tube is configured to separate gas from an inlet into a hot gas pathway and a cold gas pathway. The vortex tube includes an inlet, a cold gas outlet, and a hot gas outlet. The hot gas outlet of the vortex tube is fluidly connected to an opening defined in the first section of the pipeline. The hot gas outlet is configured to flow hot gas from the vortex tube into an interior volume of the pipeline. The compressed gas source is fluidly connected to the inlet of the vortex tube.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a divisional of and claims the benefit of priority to U.S. patent application Ser. No. 17/164,131, filed Feb. 1, 2021, the contents of which are incorporated by reference herein.

TECHNICAL FIELD

(1) This disclosure relates to a pipeline, for example, a pipeline through which fluids such as hydrocarbons are flowed.

BACKGROUND

(2) Gas hydrates are physical compounds, in which the molecules of gas are included in crystalline cells. All gases can form hydrates under different pressures and temperatures. The crystalline structure of solid gas hydrate crystals depends on the gas composition, pressure, and temperature. Formation of a gas hydrate occurs when water and natural gas are present at a low temperature and a high pressure. Such conditions can occur in oil and gas wells, and pipeline equipment.

SUMMARY

(3) This disclosure describes technologies relating to hydrate mitigation in pipelines with vortex tubes.

(4) In certain aspects, a system to prevent the formation of hydrates in a pipeline includes a heater assembly. The heater assembly includes a vortex tube mounted on an outer surface of a first section of the pipeline and a compressed gas source. The vortex tube is configured to separate gas from an inlet into a hot gas pathway and a cold gas pathway. The vortex tube includes an inlet, a cold gas outlet, and a hot gas outlet. The hot gas outlet of the vortex tube is fluidly connected to an opening defined in the first section of the pipeline. The hot gas outlet is configured to flow hot gas from the vortex tube into an interior volume of the pipeline. The compressed gas source is fluidly connected to the inlet of the vortex tube.

(5) In some cases, the heater assembly has a compressed fluid line that connects the compressed gas source to the inlet of the vortex tube. A valve can be mounted on the compressed fluid line. The valve can be configured to control a volume of compressed fluid that enters the vortex tube.

(6) Some systems further include one or more processors; and a computer-readable medium storing instructions executable by the one or more processors to perform operations. The operations include receiving signals from a temperature sensor arranged in the pipeline downstream of the opening of the pipeline; and controlling the valve mounted on the compressed fluid line to increase or decrease an amount of hot air entering the pipeline from the vortex tube.

- (7) In some cases, the heater assembly includes a fluid tubing connects the hot gas outlet of the vortex tube to the opening of the pipeline. Some heater assemblies includes a valve mounted on the fluid tubing. The valve is configured to control a volume of hot gas that enters the pipeline by the opening.
- (8) Some systems further include one or more processors; and a computer-readable medium storing instructions executable by the one or more processors to perform operations. The operations include receiving signals from a temperature sensor arranged in the pipeline downstream of the opening of the pipeline; and controlling the valve mounted on the fluid tubing to increase or decrease an amount of hot air entering the pipeline from the vortex tube.
- (9) In some embodiments, the cold gas outlet vents cold gas to the environment.
- (10) In some cases, the hot gas pathway of the vortex tube extends from the inlet of the vortex tube to the hot gas outlet of the vortex tube.
- (11) Some cold gas pathways of the vortex tube extend from the inlet of the vortex tube to the cold gas outlet of the vortex tube.
- (12) In some embodiments, the system further includes, at least one temperature sensor arranged in the interior volume of the first section of the pipeline.
- (13) Some system further include at least one pressure sensor arranged in the interior volume of the first section of the pipeline.
- (14) In certain aspects, a system to prevent the formation of hydrates in a pipeline is disclosed. The system includes a first heater assembly and a second heater assembly. The first heater assembly includes a first vortex tube mounted on a first section of the pipeline. The first vortex tube has a first inlet, a first cold gas outlet, and a first hot gas outlet. The first hot gas outlet of the first vortex tube is fluidly connected to a first opening of the first section of the pipeline. The second heater assembly includes a second vortex tube mounted on a second section of the pipeline in fluid communication with the first section. The second vortex tube has a second inlet, a second cold gas outlet, and a second hot gas outlet. The second hot gas outlet of the second vortex tube is fluidly connected to a second opening of the second section of the pipeline.
- (15) Some systems further include a first compressed gas source fluidly connected to the inlet of the first vortex tube. The first compressed gas source can be fluidly connected to the inlet of the second vortex tube. In some embodiments, the system further includes a second compressed gas source fluidly connected to the inlet of the second vortex tube.
- (16) In some cases, the system further includes at least one temperature sensor arranged in an interior volume of the first section or the second section of the pipeline. The at least one temperature sensor can be mounted on an inner surface of the first section of the pipeline, the inner surface of the second section of the pipeline, or both the inner surface of the first section and an inner surface of the second section.
- (17) Some system further include a third heater assembly having a third vortex tube mounted on an outer surface of a third section of the pipeline
- (18) In certain aspects, a method to prevent formation of hydrates in a pipeline is disclosed. The method includes receiving first temperature signals from a temperature sensor arranged in a pipeline, wherein the temperature signals indicate a temperature of a natural gas fluid flowing through the pipeline, determining, based on the temperature signals, that the natural gas fluid has a first temperature below 10° C., and in response to determining that the natural gas fluid has a first temperature between below 10° C., controlling a flow of hot gas from a heater assembly mounted to the pipeline to an interior volume of the pipeline. The flow of the hot gas prevents the formation of hydrates in the pipeline.
- (19) In some methods, controlling a flow of hot gas from a heater assembly to an interior volume of the pipeline includes controlling a valve arranged in a fluid line of a heater assembly such that a hot gas from an outlet of a vortex tube flows into the pipeline via an opening in the pipeline. In some embodiments, the fluid line is a compressed fluid line connecting a compressed fluid source of the

heater assembly to a vortex tube of the heater assembly. In some cases, the fluid line is a fluid tubing connecting an outlet of a vortex tube of the heater assembly to an opening in the pipeline.

(20) In some methods, controlling a flow of hot gas from a heater assembly to an interior volume of the pipeline includes controlling a pump rate of a pump arranged on a compressed fluid line of a heater assembly.

(21) Some methods further include receiving pressure signals from a pressure sensor arranged in the pipeline, and determining the pressure of the pipeline based on the pressure sensor.

(22) In some cases, the method further includes receiving second temperature signals from the temperature sensor, determining, based on the second temperature signals, that the natural gas fluid has a second temperature above 25° C., and controlling a valve such that valve closes or partially closes.

(23) The details of one or more embodiments of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic view of a system with a heater assembly mounted on a section of a pipeline.
- (2) FIG. 2 is a cross sectional view of the section of the pipeline shown in FIG. 1 with the heater assembly.
- (3) FIG. 3 is a flow chart of a method to prevent the formation of hydrates using a system having a heater assembly.
- (4) FIG. 4 is a schematic view of a system with a heater assembly having multiple heater assemblies mounted on sections of a pipeline.
- (5) Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

(6) A system is disclosed for eliminating hydrates or preventing the formation of hydrates in a natural gas fluid flowing in a pipeline. Natural gas is generated and transported during gas production operations. Hydrates form by combining natural gas and water under specific combinations of pressure and temperatures that may be present when transporting the natural gas in a pipeline, for example, during winter operations. The disclosed system includes a natural gas pipeline with an opening in the wall of the pipeline. A vortex tube is mounted on the pipeline and has a hot gas outlet fluidly connected to the opening of the pipeline, and a compressed gas source connected to an inlet of the vortex tube. The vortex tube separates the compressed gas entering the inlet into a cold gas flow, vented to the environment, and a hot gas flow, connected to the opening of the pipeline. The hot compressed gas mixes with the natural gas fluid flowing in the pipeline to raise the temperature of the natural gas fluid. The temperature of the natural gas fluid is raised to such a degree that new hydrates cannot form and formed hydrates are dissolved. The system and methods for eliminating and preventing hydrates from a natural gas pipeline may prevent or reduce the occurrence of hydrate plugs or blockages, thereby reducing gas production stoppages or interruptions.

(7) FIG. 1 is a schematic view of a system **100** with a heater assembly **102** mounted on a (first) section **104** of a pipeline **106**. A natural gas fluid **108** can flow in the pipeline **106** and includes natural gas, water and other compounds that are typically produced with natural gas. The system **100** is configured to heat the natural gas fluid **108** flowing in the pipeline **106** to a temperature that prevents the formation of hydrates and/or destabilizes any formed hydrates present in the pipeline **106**.

(8) The section **104** of the pipeline **106** includes a wall **110** with an inner surface **112** and an outer surface **114**. The inner surface **112** of the wall **110** defines an interior volume **118**, in which the natural gas fluid **108** is contained. The wall **110** of the section **104** extends from a first open end **120** defining an inlet **122** to a second open end **124** defining an outlet **126**. The inlet **122** and outlet **126** are fluidly connected by the interior volume **118**. Connectors **129** are mounted on the first and second open ends **120**, **124** to connect the section **104** of the pipeline to other sections of the pipeline, described in detail with reference to FIG. 4. The section **104** of the pipeline **106** is centered on an axis **116** that passes through the first open end **120** and the second open end **124**. The natural gas fluid **108** generally flows parallel to the axis **116**. The wall **110** of the section **104** of the pipeline **106** further includes an opening **128** that extends from the outer surface **114** to the inner surface **112**. The opening **128** has a diameter of about ¼ inch to about 1½ inch. The opening **128** extends in a direction perpendicular to the axis **116**, however, some openings extend at an angle relative to the axis **116**, e.g. 10°, 20°, 30°, 40°, 50°, 60°, 70°, 80°, 100°, 110°, 120°, 130°, 140°, 150°, 160°, or 170°.

(9) The heater assembly **102** includes a (first) vortex tube **130** mounted on the outer surface **114** of the wall **110** of the section **104** of the pipeline **106**. Vortex tubes include two fluid pathways that separate a gas into a hot gas flow and a cold gas flow. The vortex tube **130** includes a (first) compressed gas inlet **132**, a (first) hot gas outlet **134** and a (first) cold gas outlet **136**. The vortex tube **130** is configured to receive a compressed gas at the inlet **132** and separate the compressed gas into a hot gas pathway **138** and a cold gas pathway **140**.

(10) Hot gas in the hot gas pathway **138** exits the hot gas outlet **134**, whereas, cold gas in the cold gas pathway **140** exits the cold gas outlet **136**. The hot gas outlet **134** of the vortex tube **130** is fluidly connected to the opening **128** of the pipeline **106** by a fluid tubing **142**. The fluid tubing **142** is sealed to the hot gas outlet **134** at one end, and to the opening **128** at the opposite end. The hot gas exiting the hot gas outlet **134** of the vortex tube **130** flows into the interior volume **118** of the section **104** of the pipeline **106** to mix with the natural gas fluid **108** and increases the temperature of the natural gas fluid **108**. The fluid tubing **142** has a one-way valve **144**, e.g., a check valve or similar valve that prevents fluid in the pipeline **106** from entering the heater assembly **102**. The fluid tubing **142** also includes a control valve **146** that controls the flow of hot compressed gas into the interior volume **118** of the section **104** of the pipeline **106**. The cold gas exiting the cold gas outlet **136** can be vented to the environment or can be redirected by additional fluid tubing. The cold gas can be used for cooling welding hot spots, and/or to cool a compressor motor.

(11) The heater assembly **102** further includes a compressed gas source **148** fluidly connected to the inlet **132** of the vortex tube **130**. The compressed gas source **148** contains a compressed gas, e.g., nitrogen, or methane. The compressed gas source **148** is connected to the inlet **132** of the vortex tube **130** by a compressed fluid line **150**. A pump **152** controls the flow of the compressed gas and is arranged on the compressed fluid line **150**. The pump **152** further controls the volume of compressed gas that enters the vortex tube **130**. In some cases, a control valve, alone or in combination with the pump, controls the flow and/or volume of compressed gas that enters the vortex tube.

(12) The heater assembly includes a processor **153** operable to control the flow of hot gas into the interior volume **118** of the pipeline **106** and/or to control the flow of compressed gas into the vortex tube **130**. The processor **153** is electronically connected to the pump **152**, the control valve **146**, and a sensor arrangement **155**. The sensor arrangement **155** includes a temperature sensor **155a** and a pressure sensor **155b** in the interior volume **118** of the pipeline **106**. The processor **153** receives temperature signals containing temperature measurements from the temperature sensor **155a** and receives pressure signals from the pressure sensor **155b**. The temperature and pressure of the pipeline are indicative of an environment in which hydrates can be formed, maintained, or dissolved. The processor **153** can adjust the flow of compressed gas or hot gas to increase the temperature of the fluid flowing in the pipeline **106**. In some cases, the processor or an operator

controls an inlet pressure and the volume of the gas entering the inlet of the vortex tube by a valve. The hot gas mixes with the natural gas fluid **108** in the pipeline to raise the temperature of the natural gas fluid in the pipeline to such a degree that hydrates cannot solidify and formed hydrates are destabilized.

(13) The processor can be implemented as a computer system that includes or is connected to a computer-readable medium (for example, a non-transitory computer-readable medium) storing instructions executable by the computer system to perform the operations described here.

Alternatively or in addition, the processor can be implemented as electrical circuitry, firmware, hardware or combinations of one or more of them with the computer system

(14) The pressure sensor **155b** is electronically connected to the processor **153** and transmits signals containing pressure measurements to the processor **153**. In some systems, a temperature sensor and/or pressure sensor is mounted on each section of the pipeline. Some systems have more than one sensor per section of pipeline. Some systems have one or more pressure sensors arranged in each section of the pipeline.

(15) In some cases, the heater valve and pump of the system are mechanically controlled (e.g., not controlled by a processor). The valve arranged on the fluid tubing opens at a predetermined pressure to control hot gas into the pipeline and the pump arranged on the compressed gas line continuously flows compressed gas into the inlet of the vortex tubing.

(16) FIG. **2** is a cross sectional view of the section **104** of the pipeline **106** shown in FIG. **1** with the heater assembly **102**. The natural gas fluid **108** includes water **154** and natural gas **156** at a high pressure (e.g., 800 pound per square inch (psi) and 2000 psi) and a low temperature (e.g., 0° C. to 20° C.). Under these conditions, the combination of water **154** and natural gas **156** form hydrate particles **158**, which can collect or group together to form a hydrate plug. The heater assembly **102** controls the conditions of the interior volume **118** of the section **104** of the pipeline **106** to prevent the formation of hydrate particles **158**. The heater assembly can raise the temperature of the natural gas fluid to between 20° C. and 40° C. Some systems alter the conditions of the interior volume to dissolve any existing hydrate particles by raising the temperature of the natural gas fluid to about 20° C. to 60° C.

(17) FIG. **3** is a flow chart of a method **160** to prevent the formation of hydrates using a system having a heater assembly. The method **160** is described with reference to the system **100** illustrated in FIG. **1**, however, the method **160** may be used in any other system.

(18) The method includes flowing the natural gas fluid **108** through the first open end **120** of the section **104** of the pipeline **106** and flowing the natural gas fluid through the interior volume **118** of the section **104** of the pipe pipeline in a first direction parallel to the axis **116**. The natural gas fluid **108** includes water **154** (FIG. **2**) and natural gas **156** (FIG. **2**) at a temperature and pressure combination sufficient to form a hydrate or a temperature (e.g., hydrate formation conditions) and pressure combination sufficient to stabilize a formed hydrate (e.g., metastable hydrate conditions). Natural gas in each system is measured to generate a cooling curve that calculates the pressure and temperature ranges at which a hydrate is formed and the pressure and temperature ranges at which a hydrate is stable or unstable. A cooling curve may differ for each system due to the complexity of the natural gas system and the presence of impurities and water fluid. The pressure of the natural gas fluid **108** can be between 800 psi and 2000 psi. The temperature of the natural gas fluid **108** can be between 5° C. and 10° C. (e.g., below 15° C.).

(19) A temperature sensor measures the temperature of the natural gas fluid **108** in the section **104** and transmits the (first) temperature signal containing the measured temperature to the processor **153**. The processor **153** determines if the measured temperature is below a predetermined temperature, e.g., below 10° C., indicating that the conditions of the natural gas fluid are sufficient to form hydrates. If the measured temperature is below the predetermined temperature the processor **153** signals to the valve **146** on the fluid tubing **142** or on the compressed fluid line **150** to open and signals to the pump **152** to pump compressed gas (e.g., nitrogen gas) from the

compressed gas source **148** to the inlet **132** of the vortex tube **130**. In some cases, a pressure sensor arranged in the interior volume of the pipeline also measures and transmits a pressure signal to the processor containing a pressure measurement. The pressure measurement and temperature measurement are analyzed by the processor to determine if the conditions of the natural gas fluid are sufficient to form hydrates.

(20) The compressed gas enters the inlet **132** of the vortex tube **130** and is separated into a hot gas pathway **138** and a cold gas pathway **140**. The cold gas in the cold gas pathway **140** exits the cold gas outlet **136** to the environment. The hot gas in the hot gas pathway **138** exits the hot gas outlet **134** into the fluid tubing **142**. Hot gas then flows from the hot gas outlet **134** of the vortex tube **130** to the interior volume **118** of the section **104** of the pipeline **106** via the opening **128** in the wall **110** of the section **104** such that a temperature of the natural gas fluid **108** flowing in the interior volume **118** increases. The temperature of the natural gas fluid **108** increases to a temperature between 20° C. and 40° C. so that hydrates cannot form. This temperature increase (or second temperature of the natural gas fluid) is measured by the temperature sensor **155a**, and transmitted to the processor **153** as second temperature signals. The opening **128** extends through the wall **110** of the section **104**, perpendicular to the first direction. The method can include controlling the valve **146** or pump **152** to adjust the temperate of the natural gas fluid based on the temperature measurements. For example, if the second temperature of the natural gas fluid is high (e.g., above 25° C.), the processor may signal to the pump to stop or decrease the pump speed. If the measured temperature of the natural gas fluid is low or demonstrates a slow rate of change, the processor may signal to the pump to increase the pump speed.

(21) FIG. 4 is a schematic view of a system **170** with a heater assembly **172** having multiple vortex tubes mounted on sections of the pipeline **106**. The system **170** is substantially similar to the system **100**, however, the system **170** includes the first section **104** of the pipeline **106**, described with reference to FIG. 1, and a second section **174** of the pipeline **106**.

(22) The second section **174** of the pipeline **106** includes a second wall **176** with a second inner surface **178** and a second outer surface **180**. The second inner surface **178** of the second wall **176** defines a second interior volume **182**, in which the natural gas fluid **108** flows. The second wall **176** of the second section **174** extends from a first open end **184** defining an inlet **186** to a second open end **188** defining an outlet **190**. The inlet **186** and outlet **190** are fluidly connected by the second interior volume **182**. Connectors **129** are mounted on the first and second open ends **184**, **188** to connect the second section **174** of the pipeline **106** to the first section **104** of the pipeline **106** via a third section **192** of the pipeline **106**. The second section **174** of the pipeline **106** is centered on the axis **116**. The natural gas fluid **108** flows parallel to the axis **116** through the first, second, and third sections **104**, **174**, **192** of the pipeline **106**. The second wall **176** of the second section **174** of the pipeline **106** further includes an opening **194** that extends from the second outer surface **180** to the second inner surface **178**. The opening **194** has a diameter of about ¼ inch to about 1½ inch. The opening **194** extends in a direction perpendicular to the axis **116**, however, some openings extend at an angle relative to the axis **116**, e.g. 10°, 20°, 30°, 40°, 50°, 60°, 70°, 80°, 100°, 110°, 120°, 130°, 140°, 150°, 160°, or 170°.

(23) The heater assembly **172** is substantially similar to the heater assembly **102**, however, the heater assembly **172** includes a second vortex tube **196**, a second compressed gas source **198**, a second fluid tubing **200**, and a second compressed fluid line **202**. The second vortex tube **196** is mounted on the second outer surface **180** of the second wall **176** of the second section **174** of the pipeline **106**. The second vortex tube **196** includes a (second) compressed gas inlet **204**, a (second) hot gas outlet **206** and a (second) cold gas outlet **208**. The second vortex tube **196** is configured to receive and separate a compressed gas from the compressed gas inlet **204** into a hot gas pathway (not shown) and a cold gas pathway (not shown).

(24) Hot gas in the hot gas pathway exits the hot gas outlet **206**, whereas, cold gas in the cold gas pathway exits the cold gas outlet **208**. The hot gas outlet **206** of the second vortex tube **196** is

fluidly connected to the opening **194** of the pipeline **106** by the second fluid tubing **200**. The second fluid tubing **200** is sealed to the hot gas outlet **206** at one end, and to the opening **194** at the opposite end. The hot gas exiting the hot gas outlet **206** of the second vortex tube **196** flows into the second interior volume **182** of the second section **174** of the pipeline **106** to mix with the natural gas fluid **108** and increase the temperature of the natural gas fluid **108**. The second fluid tubing **200** has a one way valve **210**, e.g. a check valve that prevents fluid in the pipeline **106** from entering the heater assembly **172**. The second fluid tubing **200** also includes a control valve **212** that controls the flow of hot compressed gas into the second interior volume **182** of the second section **174** of the pipeline **106**. The cold gas exiting the cold gas outlet **208** can be vented to the environment or can be redirected by additional fluid tubing.

(25) The third section **192** includes a wall **216** with a first open end **218** and a second open end **220**. The first open end **218** of the third section **192** connects by a connector **129** to the second open end **124** of the first section **104**. The second open end **220** of the third section **192** connects, by a connector **129**, to the first open end **184** of the second section **174**. In this configuration, the first open end **120** of the first section **104** fluidly connects to the second open end **188** of the second section **174** via the third section **192**.

(26) The heater assembly **172** further includes the processor **153** and the second compressed gas source **198** fluidly connected to the inlet **204** of the second vortex tube **196**. The second compressed gas source **198** contains a compressed gas, e.g., nitrogen or methane. The second compressed gas source **198** is connected to the inlet **204** of the second vortex tube **196** by the second compressed fluid line **202**. A pump **214** controls the flow and/or volume of compressed gas that enters the second vortex tube **196**. The processor is electronically connected to the valves **146**, **212**, the pumps **152**, **214**, and temperature sensors **155a** to control the flow of hot gas in the first section **104** and the second section **174**. This configuration may reduce the formation of hydrates over a longer portion of a pipeline. In some cases, a control valves arranged on the first and second compressed fluid lines, alone or in combination with the pumps, controls the flow and/or volume of compressed gas that enters the first and second vortex tube.

(27) Some systems have multiple sections, e.g., three, ten, twenty, fifty, seventy, or one hundred sections, each with a vortex tube mounted on the section.

(28) In some systems, the inlet of the second vortex tube is connected to the first compressed gas source, such that the same compressed gas source provides the compressed gas to multiple vortex tubes.

(29) While the first and second sections **104**, **174** have been described as being centered on the same axis **116**, some sections are centered on different axes. The openings of the walls of each section can be perpendicular to the axis of each section.

(30) A number of embodiments have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope. Accordingly, other embodiments are within the scope of the following claims.

Claims

1. A method to prevent formation of hydrates in a pipeline, the method comprising: receiving first temperature signals from a temperature sensor arranged in a pipeline, wherein the first temperature signals indicate a temperature of a natural gas fluid flowing through the pipeline, determining, based on the first temperature signals, that the natural gas fluid has a first temperature below 10° C., and in response to determining that the natural gas fluid has a first temperature between below 10° C., controlling a flow of a hot gas, different from the natural gas fluid, from a heater assembly mounted to the pipeline to an interior volume of the pipeline, wherein the flow of the hot gas prevents the formation of hydrates in the pipeline.

2. The method according to claim 1, wherein controlling a flow of hot gas from a heater assembly

to an interior volume of the pipeline comprises: controlling a valve arranged in a fluid line of a heater assembly such that a hot gas from an outlet of a vortex tube flows into the pipeline via an opening in the pipeline.

3. The method according to claim 2, wherein the fluid line is a compressed fluid line connecting a compressed fluid source of the heater assembly to a vortex tube of the heater assembly.

4. The method according to claim 2, wherein the fluid line is a fluid tubing connecting an outlet of a vortex tube of the heater assembly to an opening in the pipeline.

5. The method according to claim 1, wherein controlling a flow of hot gas from a heater assembly to an interior volume of the pipeline comprises: controlling a pump rate of a pump arranged on a compressed fluid line of a heater assembly.

6. The method according to claim 1, further comprising: receiving pressure signals from a pressure sensor arranged in the pipeline, and determining a pressure of the pipeline based on the pressure sensor.

7. The method according to claim 1, further comprising: receiving second temperature signals from the temperature sensor, determining, based on the second temperature signals, that the natural gas fluid has a second temperature above 25° C., and controlling a valve such that valve closes or partially closes.

8. The method according to claim 1, wherein in response to determining that the natural gas fluid has the first temperature between below 10° C., controlling the flow of hot gas from the heater assembly mounted to the pipeline to the interior volume of the pipeline comprises: controlling the flow of cold gas from the heater assembly to the environment.

9. The method according to claim 1, wherein the natural gas fluid comprises water and natural gas at a pressure between 800 psi and 2000 psi.

10. The method according to claim 1, wherein hot gas is a Nitrogen gas or a Methane gas.

11. The method according to claim 3, wherein the compressed fluid source contains Nitrogen or Methane gas.

12. The method according to claim 1, wherein receiving the first temperature signals from the temperature sensor arranged in a pipeline comprises: receiving the first temperature signals from the temperature sensor arranged in the pipeline and from a second temperature sensor arranged in the pipeline.

13. The method according to claim 1, further comprising: receiving second temperature signals from the temperature sensor.

14. The method according to claim 13, further comprising: determining, based on the second temperature signals, that the natural gas fluid has a second temperature of about 40 C to about 60 C; and prompting a fluid pump to reduce a pump speed or stop pumping a compressed fluid from a compressed fluid source to the heater assembly.

15. The method according to claim 13, further comprising: receiving pressure signals from a pressure sensor in the pipeline; and determining, based on the second temperature signals and pressure signals, that hydrates in the natural gas fluid can be formed, maintained, or dissolved.

16. The method according to claim 15, further comprising: controlling a valve arranged in a fluid line of a heater assembly such that valve closes or partially closes.

17. The method according to claim 15, further comprising: prompting a fluid pump to reduce a pump speed or stop pumping a compressed fluid from a compressed fluid source to the heater assembly.
